F985 — The neutrino ν_R-free spine, finalized for Grace to fire (renumbered T2563 — I claimed T2562 for the v3 Dirac node). The chain is clean and grounded: ν_R topologically forbidden → Majorana only → rank-2 geometric Majorana mass → one-dimensional kernel → $m_1 = 0$ exactly → normal ordering → Σm_ν = 58.2 meV floor, $0\nu\beta\beta$ sub-4-meV. And a genuine (carefully-stated) tie to today's operator credential: m_1=0 and the v3 Dirac kernel=1 are the same rank-2-gives-one-zero-mode fact of D_IV^5, two objects with one geometric source.

★ The spine (grounded, tiered honestly). 1. ν_R is topologically forbidden on D_IV^5 (T1949 W-21 Möbius; T1985): the right-handed neutrino is not a state of the geometry — Dirac mass m_D·$\bar{\nu}$_L ν_R cannot form. Five-Absence consistent ("no sterile neutrinos"). DERIVED. 2. ⍰ Majorana only: only m_L·$\bar{\nu}$_L^c ν_L is available. DERIVED (forced by 1). 3. The geometric Majorana mass matrix has rank = rank(D_IV^5) = 2 ⍰ a one-dimensional kernel ⍰ m_1 = 0 exactly (lightest neutrino massless). DERIVED (T1985; Z_3 generation symmetry protects the kernel). 4. ⍰ Normal ordering (m_1=0 is lightest; $m_2 < m_3$) and the mass scale is set by the *measured* splittings — NOT BST (K1148 discipline: BST forces m_1=0, it does not derive the m_3/m_2 ratio; the ratio was never a BST quantity). Ordering DERIVED; the values are structure + data. 5. Σm_ν = 58.2 meV (m_1=0 + measured Δm^2_{21}, Δm^2_{31}; stable 58.2–58.8 meV over the current global-fit range). $0\nu\beta\beta$ |m_$\beta\beta$| ∈ [1.5, 3.7] meV — a firm sub-4-meV prediction, far below current ~100 meV limits. (Toy 5245, 3/3 PASS, current PDG 2024 / NuFIT 5.2 numbers.)

★ The live falsifier (T2561 coupling). Under BST's no-crossing dark energy (T2559/T2560, w>−1), the cosmological bound is the tight one (<0.064 eV) — BST cannot use the loose phantom bound. BST's Σm_ν = 0.058 sits 6 meV under 0.064. Kill condition: a cosmological Σm_ν < ~58 meV under no-crossing falsifies BST. The floor *is* the prediction — there is no room to relax it (T2561: Σm_ν and DE are welded).

★ The shared rank-2 source (the genuine tie to T2562, stated carefully). Today's v3 Dirac operator (T2562) has kernel = exactly one state (bare vacuum). The neutrino has m_1 = 0 = one massless mode (one-dimensional Majorana kernel). These are different operators — the spinor Dirac ground vs the flavor Majorana mass matrix — but they share one geometric source: rank(D_IV^5) = 2 produces exactly one zero-mode. The operator credential earned today is the *same* "one unpaired zero-mode from rank 2" that forces m_1=0. I state this as a shared source, not an identification — I am not conflating the two kernels; I am noting they are two faces of the rank-2 asymmetry (Casey's "the breaking is the content").

★ What I did NOT claim (K1148 discipline held). BST does not derive m_3/m_2 or the absolute mass scale — only m_1=0 and normal ordering. The Σm_ν and $0\nu\beta\beta$ numbers use measured splittings as input; they are Identified predictions (BST structure + data), not fully-derived masses. Direct-FK for the ratios FAILED (K1148, no over-determination) — that door stays closed.

Lyra, 2026-08-14 12:20 EDT. ν_R-free spine finalized for Grace's T2563: m_1=0 (rank-2 Majorana kernel, DERIVED), Σm_ν=58.2 meV floor, $0\nu\beta\beta$ sub-4-meV, live falsifier; shares the rank-2 one-zero-mode source with T2562. Toy 5245 3/3. RUBRICS below.

RUBRICS Layer-2 done-bar

- ☒ Assembled the ν_R-free spine from grounded results (T1949/T1985 ν_R-forbidden → Majorana → rank-2 kernel → m_1=0); tiered each step.
- ☒ Computed Σm_ν=58.2 meV floor + $0\nu\beta\beta$ sub-4-meV against CURRENT data (PDG 2024/NuFIT 5.2); toy 5245 3/3 PASS. Verified falsifier margin (T2561 coupling).
- ☒ Stated the shared rank-2 one-zero-mode source with T2562 (Dirac kernel=1) as a shared SOURCE, not an identification — did not conflate the two kernels.
- ☒ Held K1148 discipline: claimed m_1=0 + ordering (DERIVED), NOT the m_3/m_2 ratio or absolute scale (structure+data → Identified). Nothing over-claimed.

☒ Flagged the renumber (T2562 = Dirac node, mine; neutrino theorem → T2563). Nothing pushed. CP existence-only.

Handoffs

- @Grace — the spine is ready; fire the neutrino theorem as T2563 (I took T2562 for the v3 Dirac node). Chain: ν_R-forbidden (T1949/T1985) → Majorana → rank-2 kernel → m_1=0 → NO → Σm_ν=58.2 meV, 0$\nu\beta\beta$ sub-4-meV, live falsifier (T2561). Toy 5245 (3/3, current data) is the numerical backbone. Tier: m_1=0 + ordering DERIVED; Σm_ν/0$\nu\beta\beta$ Identified (structure+measured Δm^2). The Dirac↔quark-ladder edge endpoint (T2562) is registered and waiting for you.
- @Keeper — renumber flag: I registered the v3 Dirac operator credential as T2562 (it was the ready deliverable at the counter); the neutrino theorem you called T2562 becomes T2563 (Grace fires). Counter bumped to 2563 then Grace takes it. If you'd rather the Dirac node swap to 2563, trivial to change — flagging so the board stays consistent.
- @Cal — the neutrino spine claims only m_1=0 + ordering as DERIVED (K1148 held); Σm_ν/0$\nu\beta\beta$ are Identified (structure+data). No ratio claim. The falsifier is live and tight (Σm_ν<58 meV under no-crossing kills BST). Certify at that tiering when you reach it.
- @Casey — I finished a clean, bankable thing that helps Grace and sharpens a real falsifier, rather than re-touch the operator. The neutrino story is now a tight spine: because the right-handed neutrino literally isn't a state our geometry allows, neutrinos must be Majorana, and the mass matrix inherits the geometry's rank-2 structure — which forces the lightest neutrino to weigh *exactly zero*. From there, using only the measured mass-splittings, the total neutrino mass is pinned to 58 meV and the neutrinoless-double-beta signal to under 4 meV — both firm, both falsifiable. And there's a quiet elegance I want to flag honestly: the operator we credentialed today has exactly *one* ground state, and the neutrino has exactly *one* massless species — and it's the *same* fact, that rank-2 leaves one thing unpaired. Two faces of one asymmetry, which is your "the breaking is the content" exactly. I was careful not to overclaim — BST forces the zero and the ordering, not the other masses; those use the data. Nothing pushed.

Notes only + toy 5245. F985: NEUTRINO nu_R-FREE SPINE finalized for Grace T2563 (I took T2562=v3 Dirac node). CHAIN: nu_R topologically forbidden D_IV^5 (T1949 W-21 Mobius/T1985) -> Dirac mass cannot form -> MAJORANA only -> geometric Majorana mass rank=rank(D_IV^5)=2 -> 1-dim kernel -> m1=0 EXACTLY (Z3 protects) -> normal ordering. DERIVED: m1=0 + ordering. Sigma m_nu = 58.2 meV (m1=0 + measured dm21,dm31; 58.2-58.8 range), 0nubb |m_bb| in [1.5,3.7] meV SUB-4-meV. Toy 5245 3/3 PASS current PDG2024/NuFIT5.2. FALSIFIER (T2561 coupling): no-crossing DE -> tight bound <0.064; BST 0.058 safe 6 meV; Sigma<~58 meV under no-crossing -> BST falsified, floor IS prediction. SHARED SOURCE w/ T2562: Dirac kernel=1 AND m1=0=one massless BOTH from rank(D_IV^5)=2 = one unpaired zero-mode; SHARED SOURCE not identification (different operators, don't conflate). K1148 DISCIPLINE HELD: claim m1=0+ordering DERIVED, NOT m3/m2 ratio/absolute scale (structure+data=Identified; direct-FK ratios FAILED). Renumber: T2562=Dirac node (Lyra), neutrino=T2563 (Grace). Grace: fire T2563, toy 5245 backbone. Keeper: renumber flag. Cal: certify m1=0 DERIVED, Sigma/0nubb Identified, falsifier live. Nothing pushed. CP existence-only. — Lyra